

Mean token-type evidence closes the seed-42 neural MS2LDA annotation gap

MS2LDA research checkpoint

22 August 2026

Abstract

We tested one small correction to the accepted document-gated neural MS2LDA architecture. Its fragment and neutral-loss channel evidence was computed with a log-sum-exp, which rewards a channel merely for having a larger vocabulary. The candidate instead subtracts the log vocabulary size, making the evidence a mean on the log scale before the existing soft fragment-loss balance. The one-pass router, three-quarter document gate, seed 42, 1,000 topics, six CPU threads, data, token features, initialization, losses, and final epoch 40 all remain fixed. On leakage-filtered validation data, MAG annotation coverage rose from 51.5 to 66.3 percent, surpassing Tomotopy’s 60.7 percent. Useful high-confidence motifs—topics associated at probability at least 0.5 with compound-balanced structure overlap score (SOS) at least 0.6—rose from 148 to 185, versus 138 for Tomotopy. Mean SOS was 0.6323 and validation NLL was 8.5014, so every predeclared gate passed. The locked checkpoint was then evaluated once on test, where it retained 66.3 percent coverage and produced 234 useful motifs versus Tomotopy’s 186. This is a substantial single-seed research result, not multi-seed confirmation or evidence to replace the production backend.

1 Question and frozen comparison

MS2LDA treats spectra as documents, fragments and neutral losses as words, and recurring word distributions as Mass2Motifs [1]. The accepted balanced-gated neural checkpoint already had better completion likelihood and more useful high-confidence motifs than Tomotopy, but its broad MAG annotation coverage remained lower: 515 of 1,000 motifs versus 607. This experiment targets only that remaining gap.

The comparison fixes seed 42, 1,000 requested topics, six CPU threads, the training/validation split, vocabulary, token features, deterministic initialization, optimization schedule, losses, one-pass inference, and final epoch 40. During development the candidate run exposed named training and validation files only. Its MAG index excluded exactly the union of validation and test compound connectivity keys.

2 Failure mode and minimal correction

The accepted decoder first combined the logits within each token-type vocabulary. For type t , raw log-sum-exp evidence grows with $|V_t|$ even when the average word evidence is identical. Unequal fragment and neutral-loss vocabulary sizes could therefore bias the subsequent channel softmax for a reason unrelated to the spectrum or topic.

The candidate replaces summed type evidence with

$$e_t = \log \sum_{w \in V_t} \exp(\ell_w) - \log |V_t|.$$

The existing softmax and 25 percent pull toward equal fragment-loss mass are then applied unchanged. This preserves word rankings within each type, uses an equal neutral channel prior, and adds no learned parameter, encoder, or loss. When the option is absent or false, the exact legacy summed-evidence calculation is retained.

After count-weighted token aggregation, document evidence is applied as

$$g_d = \text{softmax}(\text{document logits}/T), \quad \theta_d = \text{normalize}(\text{token mass} \times g_d^{0.75}).$$

The forward calculation is identical in training and inference. During training, g_d is treated as fixed evidence so the shared router cannot improve the loss merely by strengthening its own gate. Multiplication preserves zero token support, and empty spectra retain the uniform fallback. Setting the type balance and gate weights to zero exactly recovers the earlier decoder and token mixture.

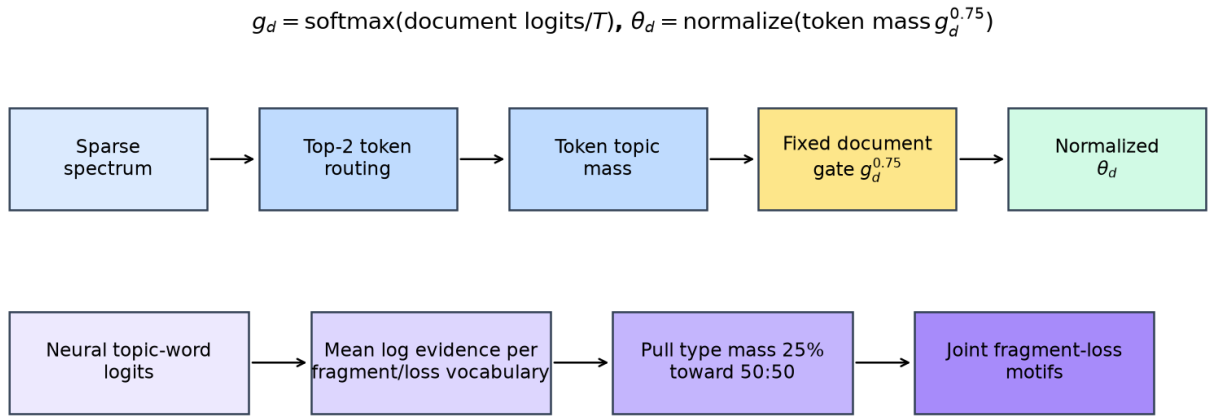


Figure 1: The candidate retains the fixed document gate and soft token-type balance, but makes fragment and neutral-loss evidence comparable by removing vocabulary-size bias before the channel softmax.

3 Paper-aligned validation

MAG annotation and SOS use validation spectra only. Library entries sharing a validation or test connectivity key are excluded before retrieval. Topic association follows the paper threshold, $\theta_{dk} \geq 0.5$. SOS is averaged over unique compounds within each topic. Eligible topics are divided into high (> 0.8), intermediate ($0.6\text{--}0.8$, inclusive), and low (< 0.6) bands.

The primary outcome is the number of high-confidence motifs with SOS at least 0.6. The unchanged checkpoint was first decoded with the new rule. Its frozen diagnostic reached 61.0 percent coverage and validation NLL 8.5042, so the one allowed full retraining proceeded. Acceptance then required all of the following:

- MAG annotation coverage of at least 60.7 percent;
- at least 148 useful high-confidence motifs;
- mean high-confidence SOS of at least 0.6318;
- validation NLL no greater than 8.5266; and
- finite, stable training and evaluation.

Runtime was descriptive rather than an acceptance gate. The candidate completed in 4,720.0 seconds, below the recorded 5,571-second context threshold.

4 Validation results

Table 1: Validation-only paper-relevant comparison.

Metric	Current neural	Mean-evidence neural	Tomotopy
Optimized motifs	515	663	607
High-confidence SOS-evaluable motifs	252	312	206
SOS high (> 0.8)	43	49	58
SOS intermediate ($0.6\text{--}0.8$)	105	136	80
SOS low (< 0.6)	104	127	68
Useful high-confidence motifs ($\text{SOS} \geq 0.6$)	148	185	138
Mean high-confidence SOS	0.6418	0.6323	0.6761
Median high-confidence SOS	0.6477	0.6364	0.6854
Model-fitting time (minutes)	59.5	78.7	76.0

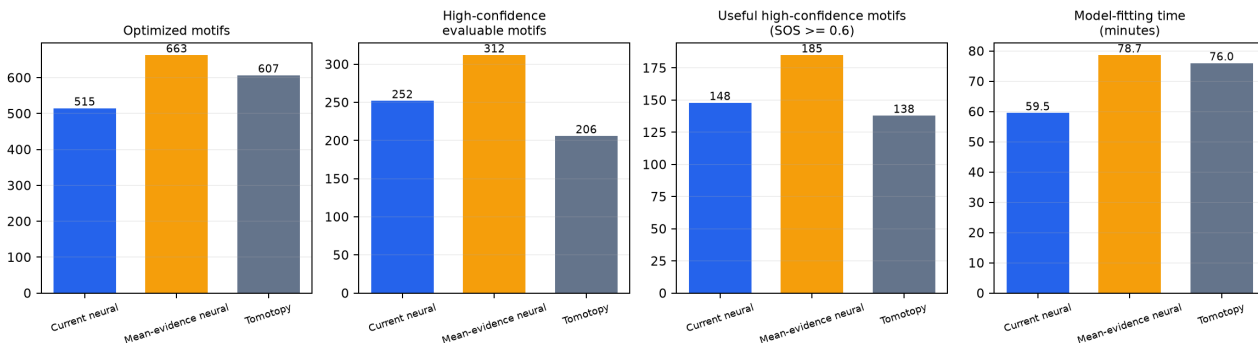


Figure 2: Headline validation outcomes and model-fitting time. The accepted model exceeds Tomotopy on both useful high-confidence motifs and broad MAG annotation coverage.

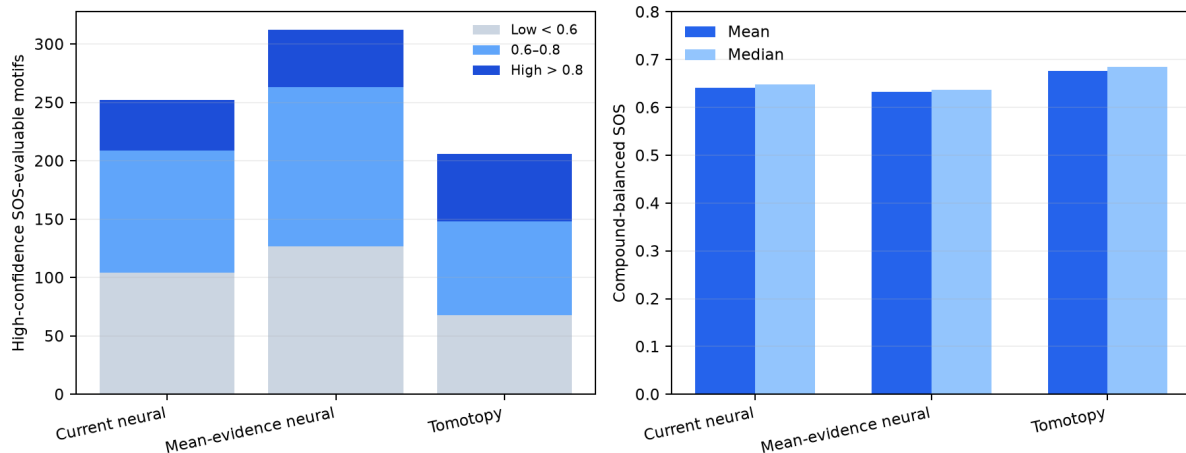


Figure 3: Validation high-confidence chemical evidence. Left: SOS bands among evaluable motifs. Right: compound-balanced mean and median SOS.

The candidate increases annotation coverage from 515 to 663 motifs, surpassing Tomotopy’s 607. It closes 160.9 percent of the remaining 9.2-percentage-point coverage gap. High-confidence SOS-evaluable motifs rise from 252 to 312, versus 206 for Tomotopy. Useful motifs rise from 148 to 185, 47 more than Tomotopy’s 138.

Candidate mean SOS is 0.6323 versus 0.6418 for the current neural model and 0.6761 for Tomotopy. Median SOS is 0.6364, versus 0.6477 and 0.6854. This is the declared tradeoff: much broader annotatability with a modest validation SOS reduction that remains above the 0.6318 floor.

Validation NLL is 8.5014, only 0.0051 above the current neural 8.4963 and below the 8.5266 safety limit. Training remains finite and stable with 321 corpus-active topics. Fitting takes 78.7 minutes, versus 59.5 minutes for the current neural reference and 76.0 minutes for the reused Tomotopy source run.

5 Single locked test confirmation

All validation checks passed before the test matrices and records were linked into the candidate run. The epoch-40 checkpoint hash was fixed, after which one neural test inference and one leakage-filtered chemistry evaluation were performed. No parameter changed after test access.

Table 2: One-time held-out test confirmation.

Metric	Mean-evidence neural	Tomotopy
Optimized motifs	663	607
High-confidence SOS-evaluable motifs	424	333
SOS high (> 0.8)	58	66
SOS intermediate (0.6–0.8)	176	120
SOS low (< 0.6)	190	147
Useful high-confidence motifs ($\text{SOS} \geq 0.6$)	234	186
Mean high-confidence SOS	0.6204	0.6369
Median high-confidence SOS	0.6135	0.6145
Held-out completion NLL	8.5226	9.7569

The test result confirms the breadth improvement. The candidate produces 234 useful high-confidence motifs versus Tomotopy’s 186 and 424 SOS-evaluable motifs versus 333. Annotation coverage is 66.3 versus 60.7 percent. Mean SOS is lower (0.6204 versus 0.6369), while the medians are nearly identical (0.6135 versus 0.6145). Candidate completion NLL is 8.5226 versus Tomotopy’s 9.7569.

6 Secondary inference context

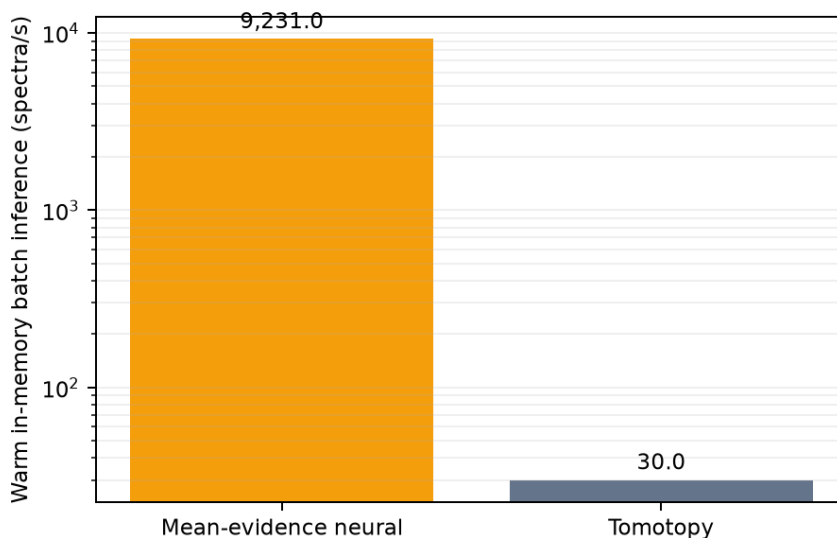


Figure 4: Matched six-thread warm in-memory batch inference on the held-out test matrices. This secondary result is not an acceptance gate or end-to-end application latency.

The accepted one-pass neural checkpoint processes 9,231 spectra/s in the fixed 128-spectrum warm batch, versus 30.0 spectra/s for Tomotopy with 100 inference iterations, a 307-fold ratio. “Warm in-memory batch inference” means that the model and batch are already loaded and one warm-up has completed. It is usable for serving or repeated batches, but excludes file loading, preprocessing, and first-call setup.

7 Interpretation

The remaining broad-coverage failure was a decoder accounting bias, not a need for a larger architecture. Mean-normalizing each token type’s evidence removes that bias without a new encoder, loss, or iterative inference step. Combined with the existing gate and soft balance, it exceeds Tomotopy on annotation coverage and useful-motif count on validation and test while maintaining much better predictive fit.

The remaining weakness is SOS strength: broader candidate coverage includes more low-SOS topics, and test mean SOS is 0.0165 below Tomotopy. This is a clear tradeoff rather than evidence that every candidate motif is superior. The result remains one fixed-seed architecture experiment. Replication across seeds is a separate confirmation stage, and no claim is made that this research checkpoint replaces the production Tomotopy backend.

8 Reproducibility

The report retains the resolved comparison summary and hashed source manifests for training, all validation references, the gate decision, the test-access record, and both test evaluations. The candidate checkpoint is fixed at epoch 40 with SHA-256:

```
639c1f37c613d908b59e3a85b7dc701e
33a3f92fd7476a3257b47298143dfbc6
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The test-access manifest confirms that selection was final before test.

References

- [1] J. J. J. van der Hooft et al. “Topic modeling for untargeted substructure exploration in metabolomics.” *Proceedings of the National Academy of Sciences*, 2016. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5137707/>
- [2] M. Torres Ortega et al. “Large-scale discovery and annotation of substructure patterns in mass spectrometry profiles.” *Nature Communications*, 2026. <https://www.nature.com/articles/s41467-026-75038-0>